

Financial Crime and Credit Risk Analytics: UK Retail and Business Bank

An end-to-end analysis of customer due diligence, AML monitoring, financial crime operating cost and credit portfolio quality, built with SQL, Excel and Power BI.

Data note: all data in this project is **synthetic** and was generated to mirror the structure of a UK retail and business bank. No real customer, employee or institutional data is used. The regulatory framework, KPI definitions and analytical methods are real and follow UK practice.

1. Business context

A mid-sized UK bank with 10 branches and roughly 3,000 customers must meet its obligations under the Money Laundering Regulations 2017 while keeping the cost of compliance under control. Financial crime teams are a fast-growing cost base, and poor onboarding decisions feed directly into credit losses later.

The bank's Head of Financial Crime and its Chief Risk Officer asked three questions:

1. Are we performing customer, business and enhanced due diligence within the standards we have set, and where do cases get stuck?
2. How much is transaction monitoring and screening actually costing us, and is that spend producing useful risk outcomes?
3. Does the financial crime risk we accept at onboarding show up later as credit risk?

2. Regulatory framework applied

Area	Requirement applied in this analysis
Money Laundering Regulations 2017 (as amended)	Risk based CDD (reg. 27-28), simplified due diligence for low risk (reg. 37), enhanced due diligence for high-risk situations (reg. 33), PEP treatment with senior management approval (reg. 35), ongoing monitoring and periodic review
Proceeds of Crime Act 2002	Suspicious Activity Reports submitted to the National Crime Agency; internal escalation from alert to SAR decision
Terrorism Act 2000	Terrorist financing typologies included in alert scenarios
Sanctions and Anti-Money Laundering Act 2018	Screening against the UK Sanctions List (OFSI), plus OFAC, UN and EU lists; true-match versus false positive disposition
JMLSG Guidance	Risk scoring by customer type, jurisdiction, product and delivery channel; source of funds and source of wealth evidence
FCA Financial Crime Guide	Governance, quality assurance of case files, management information
IFRS 9 / PRA expectations	Three-stage impairment model, probability of default, loss given default and expected credit loss

3. Data model

Ten linked tables, modelled as a star schema in Power BI.

Fact tables: CDD/KYC cases (3,202), business due diligence reviews (910), external screening events (7,691), AML alerts (6,500), loan portfolio (1,860), monthly operating cost (10,080 rows) and branch income.

Dimensions: customers (3,000), branches (10), analysts (40) and a date table created in DAX.

Method: data was loaded into SQL for cleaning and aggregation, validated in Excel against a formula-driven control sheet, then modelled and visualised in Power BI across three report pages: Due Diligence, Screening and AML, and Cost and Credit Risk.

4. Findings

4.1 Customer due diligence

Due diligence level	Cases	Average days to complete	SLA breach rate
Simplified (SDD)	566	2.5	7%
Standard (CDD)	2,219	7.1	22%
Enhanced (EDD)	391	26.3	32%

- **Enhanced due diligence takes about 10x longer than simplified** and consumed 6,592 analyst hours, which is 37% of all case handling effort for 12% of cases. High-risk customers, PEPs and customers in higher-risk jurisdictions drive this.
- **Periodic reviews are the weak point.** They average 25.3 days against 8.0 days for onboarding, because refreshing an existing high-risk file requires the same evidence as new onboarding but competes with new business for analyst time.
- **The top rejection reasons are evidential, not identity-based:** source of funds unclear (33 cases), sanctions match (32), customer withdrew (32) and adverse media (30). Identity verification itself rarely fails.
- **9% of completed case files failed internal quality assurance**, which is the metric an FCA supervisory visit would test first.

4.2 Business due diligence

Structure	Reviews	UBO not identified	Rejected
Simple ownership	575	0%	0%
Complex ownership (3+ layers or offshore)	335	28%	22%

Complex ownership structures are the single strongest predictor of a failed business onboarding. Every rejected business customer and every unidentified ultimate beneficial owner sat in the complex-structure group, which also contained the shell-company indicators: no verifiable trading address, no website and nominee-style directorships. A further 71 cases were referred into enhanced due diligence and 40 required remediation.

4.3 Screening and external research

- 7,691 screening events (initial and six-monthly rescreens) produced **680 hits**, of which **630 were cleared as false positives, a 93% false positive rate**.
- Only **5 sanctions hits were true matches**, each resulting in exit of the relationship. **45 adverse media hits were confirmed relevant** and escalated to EDD.
- Confirmed adverse media clustered in regulatory fines (61), bribery and corruption (57), fraud (54) and money laundering (50), which tells the bank where its open-source research effort is best spent.
- Average resolution time for a hit was 5.9 days, during which onboarding cannot complete. Screening noise is therefore an onboarding delay problem as well as a cost problem.

4.4 AML transaction monitoring

- 6,500 alerts generated, **91% closed as false positives**, 453 still open, and **197 SARs filed** with the NCA.
- **4,126 of 8,598 analyst hours, or 48% of all investigation effort, went to alerts closed as false positives.**
- False-positive rates were almost identical across all seven monitoring scenarios (90-92%), which indicates the thresholds, not the scenario logic, are mis-calibrated.
- High-risk customers generated **7.6 alerts each, versus 1.2 for low-risk customers**, confirming that the risk rating assigned at onboarding is predictive and can be used to prioritise queues.

4.5 Cost of financial crime compliance

Metric	Result
Total operating cost (2024-25)	£163.5m against a £149.3m budget, 9.5% over
Financial crime share of total cost	31% (KYC & Onboarding £24.1m, AML & Financial Crime £26.8m)
Cost growth 2024 to 2025	KYC +11.8% , AML +10.2% , versus 4-5% for other departments
Screening and data vendor spend	£9.6m
Group cost-to-income ratio	61.7%

Three branches, Canary Wharf, Leeds and Cardiff, overspent budget by about 15%, roughly double the 7% seen elsewhere, concentrated in staff and outsourcing. Their cost-to-income ratios (64-65%) are 4-5 points worse than comparable branches, so the overspend is not explained by higher income.

4.6 Credit portfolio quality

Credit score band	Loans	Exposure	Default rate
Excellent (740+)	395	£52.9m	0.3%
Good (670-739)	570	£75.1m	0.4%
Fair (580-669)	651	£83.4m	2.5%
Poor (300-579)	244	£38.9m	11.1%

- Total exposure **£250.3m**, NPL ratio **3.7% by exposure**, expected credit loss **£6.4m** (2.5% coverage).
- Under IFRS 9, 46 credit-impaired Stage 3 loans (£9.4m) carry £2.8m of ECL, while 69 Stage 2 loans (£12.5m) carry £1.7m. **Stage 2 is the early warning population worth managing first.**
- Concentration risk: **34% of exposure sits in London** and 84% is in property-backed lending (mortgages and commercial mortgages), so the book is highly sensitive to a UK property correction.

4.7 The link between financial crime risk and credit risk

This is the finding that matters most, because it crosses two functions that usually report separately.

AML risk rating at onboarding	Exposure	Default rate
Low	£143.4m	0.9%
Medium	£75.7m	3.0%
High	£31.2m	11.7%

Customers rated high risk for money laundering purposes default at 13 times the rate of low-risk customers.

The bank also holds **£14.6m of lending exposure to 110 customers for whom a SAR has been filed**. This is a combined conduct, credit and reputational exposure that neither the financial crime team nor the credit team currently sees in one place.

5. Recommendations

1. **Tune monitoring thresholds by customer risk band.** A 10-point reduction in the false positive rate would release roughly 450 analyst hours a year; halving it would release about 2,000 hours, worth over £50k of analyst capacity at current salary cost, and reduce the £9.6m vendor and screening spend as alert volumes fall.
2. **Separate the periodic review queue from onboarding.** A dedicated review team removes the 25-day review backlog from competing with revenue-generating onboarding, and directly addresses the 32% EDD SLA breach rate.
3. **Introduce a complex-structure triage gate.** Because 28% of complex structures fail UBO identification, routing them to a specialist business due diligence team at first contact avoids wasted standard-process effort and reduces the 22% late rejection rate.
4. **Bring the three overspending branches to the peer average.** Closing a 15% variance to the 7% peer level on their combined £51.7m cost base recovers approximately £4m over the period.
5. **Share the AML risk rating with credit risk.** Given the 11.7% default rate among high-risk customers, the rating should feed both credit decisioning and IFRS 9 staging, and any customer with a filed SAR and live lending exposure should be reviewed jointly by both teams.
6. **Strengthen quality assurance.** A 9% QA failure rate is the most likely source of a supervisory finding; targeted coaching on source of funds and source of wealth evidence addresses the leading rejection and failure reason.

6. Tools and techniques

- **SQL:** joins across customer, case, screening, alert and loan tables; `GROUP BY` aggregation for rates and averages; `CASE WHEN` for risk banding; CTEs and window functions for ranking branches and building running alert volumes.
- **Excel:** a formula driven KPI control sheet (`SUMIFS`, `COUNTIFS`, `AVERAGEIFS`, `IFERROR`) used to validate every figure produced in SQL and Power BI; pivot tables for variance analysis.
- **Power BI:** Power Query cleansing, star-schema data model, DAX measures including `CALCULATE`, `DIVIDE` and time intelligence for year-on-year cost growth, with drill-through from branch to case level.

Sample SQL, the financial crime to credit risk link:

```

SELECT  c.risk_rating,
        COUNT(DISTINCT l.loan_id)                AS loans,
        SUM(l.exposure_gbp)                      AS exposure_gbp,
        ROUND(100.0 * SUM(CASE WHEN l.default_flag = 'Y'
                                THEN 1 ELSE 0 END) / COUNT(*), 2) AS default_rate_pct
FROM    loan_portfolio l
JOIN    customers c ON c.customer_id = l.customer_id
GROUP BY c.risk_rating
ORDER BY default_rate_pct DESC;

```

Sample DAX, the AML false positive rate:

```

False Positive Rate =
DIVIDE(
    CALCULATE( COUNTROWS( AML_Alerts ),
                AML_Alerts[outcome] = "Closed - False Positive" ),
    CALCULATE( COUNTROWS( AML_Alerts ),
                AML_Alerts[status] = "Closed" )
)

```

7. Skills demonstrated

Customer due diligence and EDD, business due diligence and UBO identification, sanctions, PEP and adverse media screening, open-source research, transaction monitoring and SAR decisioning, UK AML regulation (MLR 2017, POCA 2002, SAMLA 2018, JMLSG), credit risk and IFRS 9 staging, cost and variance analysis, SQL, Excel, Power BI and DAX, translating analysis into operational recommendations.